THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 9 & 10: **Cell: The Building Block of Life, Tissues in Action** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The statement ‘new cells arise only from pre-existing cells’ was given by: 1

(a) Brown **(b)** Rudolf Virchow **(c)** Hooke **(d)** Schwann

2. If a cell is 200 µm in size, in millimetres it is: 1

(a) 0.02 mm **(b)** 0.2 mm **(c)** 2 mm **(d)** 20 mm

3. A potato piece placed in 20% salt solution will: 1

(a) swell **(b)** dissolve

(c) stay unchanged **(d)** shrink as it loses water

4. The cell wall in plants is mainly made of: 1

(a) chitin only **(b)** lipid **(c)** cellulose **(d)** protein

5. The functional segments of DNA are called: 1

(a) ribosomes **(b)** plastids **(c)** genes **(d)** chromatin

6. Growth in the length of plant roots and shoots is due to: 1

(a) intercalary meristem **(b)** apical meristem

(c) lateral meristem **(d)** epidermis

7. The annual rings in a tree trunk help to estimate its: 1

(a) root length **(b)** height **(c)** mass **(d)** age

8. Loss of water vapour through stomata is called: 1

(a) respiration **(b)** excretion **(c)** transpiration **(d)** osmosis

9. Cells with unevenly thickened corners that give flexibility to stems and leaves form: 1

(a) xylem **(b)** sclerenchyma **(c)** collenchyma **(d)** parenchyma

10. Food prepared in leaves is transported to other parts by: 1

(a) stomata **(b)** xylem **(c)** phloem **(d)** cork

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Name the chief constituent of the plant cell wall. Give one use of it in our diet. 2

12. What is a prokaryotic cell? Give an example. 2

13. Differentiate between parenchyma and collenchyma. 2

14. Why is sclerenchyma strong? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Why do plants have chromoplasts and leucoplasts in addition to chloroplasts? 3

16. Why is cork formed in older stems? State its function. 3

17. Describe the structure and function of a plant vacuole. 3

OR

A plant wilts when not watered. Explain with reference to cells. 3

18. Name the types of epithelial tissue with one example each. 3

OR

How are epithelial tissues adapted to their functions? Give two examples. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Describe plastids and their types. 5
(ii) State two features that resemble bacteria.
(iii) Rohit says roots do not have plastids. Is he correct?

OR

Case study: Onion root tip cells show different shapes of chromatin under the microscope. 5
(i) Why are root tips used?
(ii) What process is seen?
(iii) Name its types.
(iv) What is the use of this process in the body?

20. (i) Name the types of epithelial tissue. 5
(ii) Relate structure to function in two examples.
(iii) State the general functions of epithelium.

OR

(i) Describe the types of connective tissue. 5
(ii) Differentiate between a tendon and a ligament.
(iii) Why is blood a connective tissue?

END OF QUESTION PAPER — Total: 40 Marks

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